

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

**A MACHINE LEARNING APPROACH TO TELECOM CHURN PREDICTION AND
MANAGEMENT: ANALYSIS AND MODELING OF EFFECTIVE ML CLASSIFIERS
FOR DECISION MAKING**

BY

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**A THESIS SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE,
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CHAPTER 1

1.0 Introduction

The introduction to the research report is included in this chapter. The main purpose of this paper is to conduct a study on a machine learning approach to Telecom churn prediction and management by analysing and modeling the most effective Machine Learning (ML) classifier for decision making. The background to the study, statement of the problem, purpose of the investigation, research questions, and justification of the study, as well as the scope of the study and how the work is organized, are all discussed in this chapter.

1.1 Background of the study

In the modern world, telecom businesses produce a massive amount of data at an incredibly fast rate. Numerous telecom service providers are vying for market share by offering better services. Customers have a variety of options, including better and more affordable services (Ullah et al.,2019). The ultimate objective of telecom businesses is to maximize profit while surviving in a cutthroat industry. According to Qureshi et al (2013), when a sizable portion of customers are dissatisfied with the services provided by any telecom company, this is known as customer churn. Customers start switching to other service providers as a result of the service relocation. Churning can occur for a number of causes. Prepaid users are not tied to a service provider like postpaid customers are, and they are free to switch at any moment. Churning also negatively affects a company's reputation as a whole, which damages its brand. A devoted consumer brings in a lot of money for the business and is rarely impacted by rival businesses.

In order to keep their important clients and improve the management of their Customer Relationship Management (CRM), telecom operators must be able to predict churn. Retaining current clients is the CRM's most difficult task. Customers have the choice to change service

providers due to the crowded and competitive market. Since it is less expensive to identify and keep existing customers than to find new ones, telecom companies have developed systems to do so. This is because keeping existing clients can cost up to five or six times as much as investing in ads, labor, and concessions. Finding the current churn customers, which can help turn the issue around, requires little care. An accurate and high-performance model for identifying clients who churn must be developed in order to meet the criterion of customer retention. To prevent customer loss and offer strategies for customer retention, the proposed model should be able to recognize consumers that churn and then determine the causes of churn. It should also use methods to foresee when a circumstance like this will occur in the future. There are numerous data mining and machine learning technologies available that may be used to evaluate such data as a result of recent developments in the field of big data. These methods examine the data to determine the causes of client churn.

These strategies can be used by CRM to increase their profit. Additionally, it can be utilized to create retention methods that lower the ratio of customers that are going to churn. By accurately detecting customers' wants through the use of data mining tools, the CRM can help a business achieve its goal of client retention. Data mining is the technique of determining customer turnover behavior using patterns that have been retrieved from the data. Numerous terms, including business intelligence, predictive modeling, knowledge discovery, and predictive analytics, are used to refer to data mining. One of the dimensions of CRM is data mining, and the CRM analyst should combine the three key dimensions of decision makers, data, and data mining.

Due to extensive reforms in the telecom industry over the past decade, which led to tremendous saturation and increased competition due to increased competition, assessing the loss suffered due to value churning consumers and maintaining current revenue of companies. The origin and

frequency of churn are difficult to anticipate. The quality component is the primary cause of many issues with customer development, including service quality, network coverage, load faults, billing, charges, technologies, etc. Customers can compare service quality and benefits with those of other service providers who offer comparable services thanks to these service quality factors. Despite the potential that it is not about acquiring new consumers, the telecom business can perform very well and handle existing customers. The regular client prediction rate in the telecom industry is typically calculated at 2%, which equates to a total annual loss of over \$100 billion (Reichheld & Sasser, 2004). The cost of introducing new consumers is 5 to 6 times higher than the cost of retaining existing customers, while predicting customer churn is 16 times less expensive (Manzano-Machob, 2013). Profit improves from 25% to 85% when the churn rate is reduced by 5% (Ph, 1999). Technological advancements have made it easier for businesses to comprehend alternative aggressive strategies to maintain a high degree of client churn. The main goal is to locate the value churn customer using a significant amount of telecom data, according to existing studies. Existing models have a number of flaws, though, and these provide significant barriers to solving this issue in the actual world. In the telecom industry, a lot of data is produced, and a lot of it has missing values, which causes the prediction models to perform poorly. In order to address these problems, data preparation techniques are modified to eliminate noise from the data, which is successful for a model to accurately categorize the data and enhance performance. While some information-rich characteristics are taken into consideration while developing models, feature selection has been employed in literature (Huang & Kechadi, 2013). Predictive model performance suffers in a variety of domains where statistical approaches are often applied. Benchmark datasets (Pendharkar, 2009) that are unable to give the real representation of the data and are not helpful to decision-makers have been used in previous studies to validate models. In order to overcome this restriction, the best classifier is chosen for retention after many methods are applied to the same

dataset. Automated churn prediction and retention models can be created with the use of the intelligent mechanism. The selection of features is yet another significant issue with current models. There are various factors that influence why a consumer or a group of customers churns. Without knowing the specific causes and contributing factors, a customer who is churning is simply labeled as a churner. Churners exhibit varying behavioral patterns; thus, they are all not addressed equally. Different types of clients have different churn rates. Such a prediction model is required so that retention strategies, such as various promotions for various churn groups of customers depending on their churn factors, can be provided. Information Gained and Correlation Attributes Ranking Filter feature is employed in the selection procedures and chooses the best features to create both results in order to overcome the aforementioned restrictions.

A suggested a churn prediction model in this work that makes use of multiple machine learning algorithms will be used.

1.2 Statement of Problem

According to Van den Poel & Lariviere (2014), a vast amount of data is available as a result of the rapid expansion of the data transmission network and improvements in information technology.

Because of the increased market competition, businesses now spend more time encouraging their existing customers to stick with them than acquiring new ones. Van Den Poel & Lariviere (2014), who conducted research on the significance of the economic value of client retention, have provided justification for this. Customer churn is a critical factor in the survival and growth of the telecom business because customers are the primary source of profit. The development of tools for predictive modeling and classification of different systematic task is key to decision making. There are likely to be tremendous rates of research in data mining and their applications in customer

churn, but still it is an active research field and researchers are searching for more accurate solutions.

1.3 Aims and Objective

The main objective of this study is to propose and analyse the most efficient and effective machine learning classifier for customer churn prediction to enhance decision making.

The above aim of the research will be accomplished by fulfilling the following research objectives:

1.4 Research Objectives

1. To evaluate some selected machine learning techniques used in customer churn prediction
2. To find the most appropriate machine learning technique for identifying customers churn.
3. To propose a machine learning model that is most capable of identifying customer churn rate in the telecom sector.

1.5 Research Questions

1. Which machine learning techniques are used in evaluating customer churn prediction?
2. Which of the techniques is most appropriate for identifying customers churn rate?
3. Is there a machine learning model that is most capable of identifying customers churn in the telecom sector?

1.6 Justification of the study

The telecom industry is growing to become one of the biggest in the world. Technological developments and an ever-growing operator base have increased the degree of competition in the telecom industry. Customers now have more options, which has raised the churn rate. Customers' behavior has changed as a result of the additional options available, raising their expectations of

the service providers they use. Consumers of the new generation are more demanding than those of previous generations, and they have access to a wider selection of options, which causes customer churn.

1.7 Methodology

In this study, customer data sample from a telecom service will be used to analyze the customers that churn or are most likely to do so. The task at hand involves analyzing data on the characteristics and behavior of clients who have either churned or remained loyal. The data used in this exercise will be sourced from the OpenML public repository and pertains to the telecommunications industry. The aim is to use Machine Learning techniques to develop models that accurately predict whether a client is likely to churn or not. This is a supervised classification problem, and the success of the models will be evaluated based on their accuracy and performance. Ultimately, the goal is to identify the best model for telecom businesses. Three machine learning models will be utilized to analyze the telecom data. These models are Random Forest, Gradient Boosted, AdaBoost and Logistic Regression ML Classifiers. The reasons for choosing these algorithms are because of their inordinate ability to handle complex heterogenous datasets, produce interpretable results as well as examining huge datasets and also predict accurately datasets comprising a variety of factors such as usage patterns, financial information, and account history, it is possible to identify which customers are most likely to leave.

Key to the aim of this research is to attempt to analyze and recommend the best machine learning classifier for predicting customer attrition in order to improve decision-making. And consider how the difficulties associated with using these machine learning techniques differ from the methods themselves.

CHAPTER 2

LITERATURE REVIEW

2.0 Introduction

In an effort to address the customer churn problem in the telecom sector, this research compares the performance of the Random Forest Classifiers, Gradient Boosted, AdaBoost and Logistic Regression ML Classifiers. General overview of telecom churn, machine learning approaches to telecom churn, Analysis of machine learning classifiers, and related literature are covered in this chapter. A summary of earlier study on the subject concludes the chapter. Machine learning in telecom churn is a relatively new area of study, and although significant work has been done in this area over the past few years, the majority of the earlier research has been focused on developing customer retention models, churn prediction and management strategies, intelligent predictions, etc. Meanwhile, not much work has been done in the area of machine learning although majority of earlier works focused on churn retention mechanism and strategies and also the main causes of customer churn and its management. Although there may be technical problems associated with telecom churn, the machine learning resolution provides an overview of the entire customer service structure in the telecom sector. The evaluation of relevant materials regarding customer churn and prediction is presented in the sections that follow.

2.1.1 Background To Telecom Churn Problem

The pace at which customers transfer telecom service providers is referred to as telecom churn, also known as customer churn or attrition. Because customers have many options and are frequently seduced by rivals that offer better pricing, features, or customer service, churn is a concern in the telecom sector. Significant resources are frequently used by telecom firms to keep customers and lower turnover. Over the years, there has been a lot of research on telecom churn, examining the many causes of churn and mitigation tactics. A summary of some of the major conclusions include customer satisfaction, switching cost, brand image, service quality and customer loyalty programs. Babakus & Yavas (2004), investigated the influence of

management commitment to service quality on service recovery performance as triggered by frontline employees on customer satisfaction and company growth. The study used data collected from frontline telecom service employees from both public and private service providers in Turkey. A conceptual model of service recovery was created and consisted of Appraisal, Emotional Response and Behaviour as its core components. A five-item empowerment scale was used to assess the performance, ranging from 5 (strongly agree) to 1 (strongly disagree). The research was successful at expanding the knowledge of the role of management commitment to service quality in recovery from service failures and identifies the underlying process as well. However, the constructs in this study were developed by responses from the same participants. Srinivasan et al., (2002) also investigated the antecedents and consequences of customer loyalty in an online business-to-consumer (B2C) context. The research employed the development and preliminary testing of an instrument containing multiple-item scales for its constructs. A dataset of the market research company's list of internet customers was randomly selected and 5,000 customers were used. Questionnaires containing various scale items were developed and respondents attended to the first 25 sets leading to refinement of the remaining questionnaires. The responses were validated with a three-way segmentation of the responses of the dataset and then tested using reliability and unidimensional as its metrics. The results indicated that e-retailers can establish early warning systems based on continuously measuring customer perceptions for the 8Cs. The limitations to the study include the model's inability to consider individual level variables.

One study by Khaled Al-Momani and Saif Al-Mashaqbah (2021) found that customer satisfaction is negatively affected by churn in the telecom industry. The study also found that service quality and customer support are critical factors in determining customer satisfaction.

Another study by Ali Adnan Ibrahim and Saad T. Abdul-Rahman (2019) explored the impact of churn on customer satisfaction in the telecom industry in Iraq. The study found that churn has a negative effect on customer satisfaction, and the level of customer satisfaction is higher among customers who have not churned.

In a study by Ahmed Abu-Shanab and Ahmed Abu-Shanab (2019), the authors examined the impact of customer churn on customer satisfaction in the telecom industry in Jordan. The study found that churn has a significant negative impact on customer satisfaction, and that customer satisfaction is positively affected by factors such as perceived value, service quality, and customer loyalty.

A study by K.K. Ramachandran and A. Palanivelu (2017) investigated the relationship between churn and customer satisfaction in the Indian telecom industry. The study found that churn has a negative impact on customer satisfaction, and that customer satisfaction is positively affected by factors such as network quality, pricing, and customer support.

Finally, a study by Adebayo Felix Adekoya (2016) explored the relationship between churn and customer satisfaction in the Nigerian telecom industry. The study found that churn has a significant negative impact on customer satisfaction, and that factors such as network quality, pricing, and customer support are critical in determining customer satisfaction.

In conclusion, the literature suggests that telecom churn has a negative impact on customer satisfaction, and that factors such as service quality, customer support, perceived value, and customer loyalty are crucial in determining customer satisfaction in the telecom industry.

2.1.0 Telecom Churn Prediction Models

For telecom operators, customer retention expenses are four times lower than new customer acquisition expenses. Because of this, churn prediction has become a crucial issue for significant service providers worldwide. This is why substantial resources are being invested more and more in the creation of fresh anti-churn tactics. Lu et al (2012), conducted a real-world research on telecom churn prediction and proposed the use of AdaBoost to enhance customer prediction. The dataset was trained repeatedly and the weighted base of the training dataset of the classifier was measured. The advantage of the investigation was the inclusion of the “Implementation Zone” within which the company is able to perform retention actions. The

data set was obtained from a telecommunications provider that served millions of active mobile subscribers at the time. The training set consisted of 7190 randomly selected customers, 678 of whom were churners and 6512 of whom were not. The model was cross-validated using a testing set of six-month customer data that was gathered in 2011. The concept allowed for the creation of an Implementation Zone where the most likely-to-churn consumers could be targeted for retention initiatives. In this study, the base learner was employed directly by a boosting algorithm that separated the training data set according to how challenging it was to fit a base learner and created a unique prediction model for each identified cluster. The paper identified a further improved performance and other classification methods as well as other techniques and business point of view. The researchers proposed an improved hybrid classifier or building a multiple boosted regression model using the sampling techniques. Another authors, Umayaparvathi & Iyakutti were of the view that businesses were becoming highly saturated especially telecommunication service providers as a result of the numerous vibrant service providers in the sector. They therefore premised their research on the cost of acquiring new customers as compared to retaining existing customers. This study claims that a number of data mining methods have been put forth in the literature to predict churners from heterogeneous customer records. The article lists the many types of customer data that are accessible in open datasets, as well as the performance measures and predictive models that have been used in the literature to anticipate churn in the telecom business. Each mobile subscriber's historical call data as well as all other personal and professional data that the service provider maintains was used as the data set for this research. Labels were also provided for the training phase in the form of a list of churners. The model was able to predict the list of churners from the real dataset that contains no churn label after being trained with the highest degree of accuracy. These researchers suggested a five-phase model for predicting client attrition. Preprocessing the input customer records came first, then feature extraction for churn model development, model construction with various classifiers and cross validation, prediction accuracy calculation, variable importance report identification, and finally providing CRM executives with customer retention policies.

2.2.1 Machine Learning Churn Prediction

Liao et al., (2012) depict phases of a general churn prediction model such as data collection, preparation, classification and prediction. It also describes that identifying the right grouping of variables has significant influence in improving the percentage of true predictions (TP). A churn prediction model was proposed by Liao et al., which works in 5 steps: i) problem identification; ii) dataset selection; iii) investigation of data set; iv) classification; v) clustering, and vi) using the knowledge. It seems to be a complete model. According to them classification techniques are used for distinguishing Churners. Clustering is used for model evaluation. For classification, they explored Decision Tree, Support Vector machine and Neural Networks. And for clustering Simple K-Means was used. It concluded that SVM was the best among the three methods in distinguishing churners from non-churners. (Xu, 2011), noted that building an effective customer churn prediction model using various techniques has become a significant topic for business and academics in recent years. These authors proposed a decision tree based Random Forest method for feature extraction. From the original data with Q features, exactly N samples with $q < Q$ were randomly selected for each tree to form a forest. The number of trees was dependent on the number of features randomly combined for each decision tree from the whole Q features. Development of a predictive model based on data-centric approach for detecting the early warning signs of churn, and for developing a Churn Score to identify subscribers who are likely to end their relationship with the company was discussed in Huang et al. (2012), where the customer's calling pattern played the major role in predicting churn.

Lu et al. (2014), attempted to solve the telecom churn problem by proposing the use of booting to enhance customer churn prediction. The paper asserts that, unlike other research works that uses boosting as a method to improve on the accuracy of a given basis learner, this paper would try to separate the algorithm into two clusters based on the weight assigned by the boosting algorithm. The paper used the AdaBoost Algorithm to train the data set based on the weight of the base classifier and hence synthesizing these base classifiers. The paper employed a single logistic regression fitted on an entire training set to compare the results of the experiments. The results of

the evaluation showed that in a customer churn data, where the weight of the data is skewed as a result of the AdaBoost separation, there is the opportunity to define a high-risk customer group. The paper identifies extensive work from both the techniques and the business aspects. It suggests further that, building a multiple boosted regression algorithm model using a sampling technique could improve predictions and decision making. The paper identifies dissatisfaction as a major player in customer churn and therefore could be tackled on the business side.

According Bock & Poel (2012), a method for classification that satisfies the two criteria of strong classification performance and high model interpretability should be used in order to develop a successful customer churn prediction model. The paper sought to reconcile performance and interpretability in customer churn prediction using ensemble learning based on generalized additive models using GAMensPlus on six real life datasets as well as case studies. The techniques used were the generalized additive models (GAM & GAM as ensemble), bagging, random forests, random subspace method and logistic regression. They used dataset from Bank, DIY, supermarket, telecom and mail order garments. They metrics such as Accuracy, AUC & TDL for the measurement of performance. Featured selection was used as sampling and 5 x 2 cross validation was performed. Then also, non-parametric Friedman test followed as well as Holm's procedure. In the experimental comparison on data sets of the six real-life churn prediction projects, the competitive performance of the proposed algorithm over a set of well-known benchmark algorithms was demonstrated in terms of the four-evaluation metrics and results were as follows. First, it demonstrated how one can determine the relative relevance of churn predictors in relation to the criterion used to assess the accuracy of the model predictions using the generalized feature importance scores. Second, GAMensPlus was shown to be able to recognize nonlinear correlations between predictors and churn probability.

Verbeke et al., (2012) also attempted to study new insights into churn prediction in the telecommunication sector a profit driven mining approach. According to the authors, customer retention campaigns can be more effective and turnover costs can be decreased by identifying the customers who are most likely to

leave. This is the goal of customer churn prediction models per the paper. Churn prediction models are often evaluated using statistically based performance indicators, which leads to inefficient model selection even though cost reduction is their primary goal. The goal was to measure the optimal model and fraction of customers to include and benchmarking experiment evaluating various classification techniques in telecom sector. The algorithms used are Decision trees, logistic model tree, bagging, boosting, random forests, nearest neighbors, neural networks, rule induction techniques, logistic regression, naïve Bayes, Bayesian networks, SVM Logistics regression, classification trees and bagging. The dataset used was from the Telecom and the measure of performance was the Maximum Profit, AUC and TDL. The authors used oversampling, fisher score, non-parametric Friedman test followed by the post-hoc Nemenyi test. According to the experimental findings, only a limited set of variables was necessary to accurately forecast churn, and oversampling typically has little to no impact on performance. Finally, it is discovered that the performance of a large number of classifiers is comparable.

Ballings & Poel (2012) investigated customer event history for churn prediction. According to the study, while most studies in predictive churn modeling aim to improve models by data augmentation or algorithm improvement, this study focuses on another dimension: time window optimization with respect to predictive performance. The techniques used were the Logistic regression trees and bagging. The dataset used was from the Newspaper Customer Feat. The performance metrics used was AUC with no sampling with Logistic regression, DeLong TEST PCC test. The findings indicated that predictive performance only slightly improves after the fifth additional year, indicating that the corporation in this study can delete 69% of its data with almost any loss in predictive performance. The practical result was that analysts could significantly lessen the workload associated with data, including data preparation, storage, and analysis. This is especially helpful in the era of big data, when it is crucial to reduce computing complexity.

2.2 Customer Attributes and Datasets

Customer behavior information provides useful insights into potential decisions. Churn prediction models, for instance, identify clients who cease using a service or product. Consumer behavior, as defined by Engle

et al. (2015), is the action and decision-making process of people who purchase products and services for their own consumption. Forecasting consumer behavior is crucial because it helps businesses identify their target market by differentiating consumers from one another. It also helps businesses retain potential customers, redesign marketing campaigns, predict market trends, foster competition, and innovate both their current products and new ones. In contrast to the customer-to-business context, where it has received little research, business-to-business contexts place a lot of focus on customer churn prediction. According to Figalish et al. (2019), customers' frequency is frequently significantly lower in B2B firms, but their transactional value is typically much larger. Because of this, a corporation is very interested in a single customer, and attrition can have a much bigger impact. This demonstrates that predicting client churn in the B2B space is appropriate. Contrarily, due to their complex configurations, B2B environments typically cannot be merged with B2C systems' measures.

2.3 Customer Churn Performance Metrics

The identification of why customers give up their relationships has been focused of marketing research for the past few years (Hashmi et al., 2013). Due to the enormous growth of customer related data and call detail data collected and maintained by the companies in the recent years, more sophisticated metrics have evolved to describe customer behaviour and better understand how behavioural characteristics can be linked to customer retention and firm performance (Umayaparvathi &Iyakutti, 2012). An overview of socially grouped users and their behavioural pattern are elaborately identified by Shabba et al. (2012). They also explored the impact of centrality features among customers. It concludes that when a customer in a group leaves the network, there is a high probability of others in that group to leave the network. It classifies the feature variables in two types: i) dependent variables (call duration of in-degree and out-degree), ii) independent variables (social forum like services and the customer's involvement in the forum). The level of customer's active participation was used as the measure of probability of churn. Customers who are members of multiple community forums are at high risk than those who are members in less number of forums.

2.3 Related Literature

Coussement et al. (2017), studied the comparative analysis of data preparation algorithms for customer churn prediction using case study in the telecommunication's decision support system. The goal was to study the effects of data preparation techniques on predictive performance in the telecommunication industry. According to the authors, data preparation is a process that aims to convert independent (categorical and continuous) variables into a form appropriate for further analysis. Their study sought to examine data-preparation alternatives to enhance the prediction performance for the commonly-used logit model. The algorithms tested were Logistic regression, Bayesian network, Naïve Bayes, Decision tree, neural network, random forest, SVM and SGB. The dataset consisted of 30,104 customers from the telecom industry with 956 features. The study was conducted in a churn prediction modeling environment, benchmarks an optimized logit model against eight state-of-the-art data mining techniques that were used as standard input data, including real-world cross-sectional data from a large European telecommunication provider. The results led to following conclusions. (i) Analysis shown that the data-preparation technique they chose actually affected churn prediction performance; It was also found that improvements of up to 14.5% in the area under the receiving operating characteristics curve and 34% in the top decile lift. (ii) The enhanced logistic regression also was competitive with more advanced single and ensemble data mining techniques.

Ullah et al., (2019) attempted to solve the customer churn problem by proposing a churn prediction model that uses classification, as well as, clustering techniques to identify the churn customers and provides the factors behind the churning of customers in the telecom sector. A number of experiments were carried out on two different labelled datasets. The experiment was performed on a wide range of algorithms that included Random Forest, Decision Stump, J48 and Random Tree with 10-fold cross-validation. Other than analyzing individual algorithms, hybrid algorithms were also selected for experimentation. This includes AdaboostM1C, Decision Stump and Bagging as well as Random

Tree algorithms. Additionally, the prediction model was also tested using the Bayes algorithm which too lied in eager learning class and performs better on larger datasets consisting of millions of records. Per the authors narration, this could be used for real-time prediction, multi-class prediction, text classification, spam filtering, sentiment analysis, and recommendation system. The classification algorithms, Random Forest (RF), Artificial Neural Networks (ANN), decision tree, C5, Multilayer Perceptron (MLP) and Logistic Regression (LR) were also used in the simulation experiment. The classification process was performed using WEKA 3.8 toolkit. The performance evaluation was done using Accuracy, Recall, Precision and f-1 measure. Different classification techniques are applied to two churn datasets using WEKA toolkit. Results were measured in terms of correctly classified, incorrectly classified and time to build tree from the two datasets as mentioned. The results indicated that the accuracy of all algorithms with 10-fold cross validation. It was observed that the Random Forest and J48 outperformed other techniques for both datasets and correctly classify the data with 88.63% and 88.58% accuracy respectively. Decision Stump and Random Tree turned out to have very low accuracy rate with 70.98% and 84.34%, respectively. The study was successful at providing guidelines on customer retention for decision makers. The study identified artificial intelligence techniques for predictions and trend analysis.

In order to address the telecom churn issues, the telecom industry is currently facing, this study will investigate the various machine learning models and classifiers capable of identifying customer that are likely to churn using a dataset from the telecom industry. This study expands upon the churn prediction model using random forest: analysis of machine learning techniques for churn prediction and factor identification in telecom sector (Ullah et al., 2019).

CHAPTER 3

METHODOLOGY

3.0 Introduction

This chapter discusses the research methodology. It discusses the methods and procedures used to conduct the study. Additionally, it details the testing procedures, network environment design, traffic generation, data gathering strategy, ML training and testing model employed, and experimentation procedure.

In this study, the churn-bigml-20.csv dataset will be used. Three machine learning models—Logistic Regression, Naïve Bayes, and AdaBoost Algorithms will be utilized to model the trained dataset on these three algorithms.

Every algorithm is modelled and tested on the dataset to assess its performance. The models will be cross-validated on the dataset but in a different proportion.

The main steps in the development process are listed.

1. Data cleaning: the dataset will be assessed thoroughly to remove null and incomplete data.
2. Data importation: the data will be imported into the python jupyter environment
3. The clean data is assessed with feature selection
4. The ML specific library is then imported and initiated
5. The first ML model is built and the sample is trained
6. The trained set is then tested on the model and the performance is recorded.
7. The performance such as Accuracy, Confusion Matrix and the others are recorded for all the algorithms

8. A cross-validation is carried out on the data and the result generated.

9. In order to achieve the goals of this study, several experiments will be conducted, and the results of each machine learning technique will be examined and analyzed.

This study aims to address the problems identified in the problem statement by using the aforementioned strategies. And consider the variations in the algorithms and difficulties associated with applying these machine learning techniques.

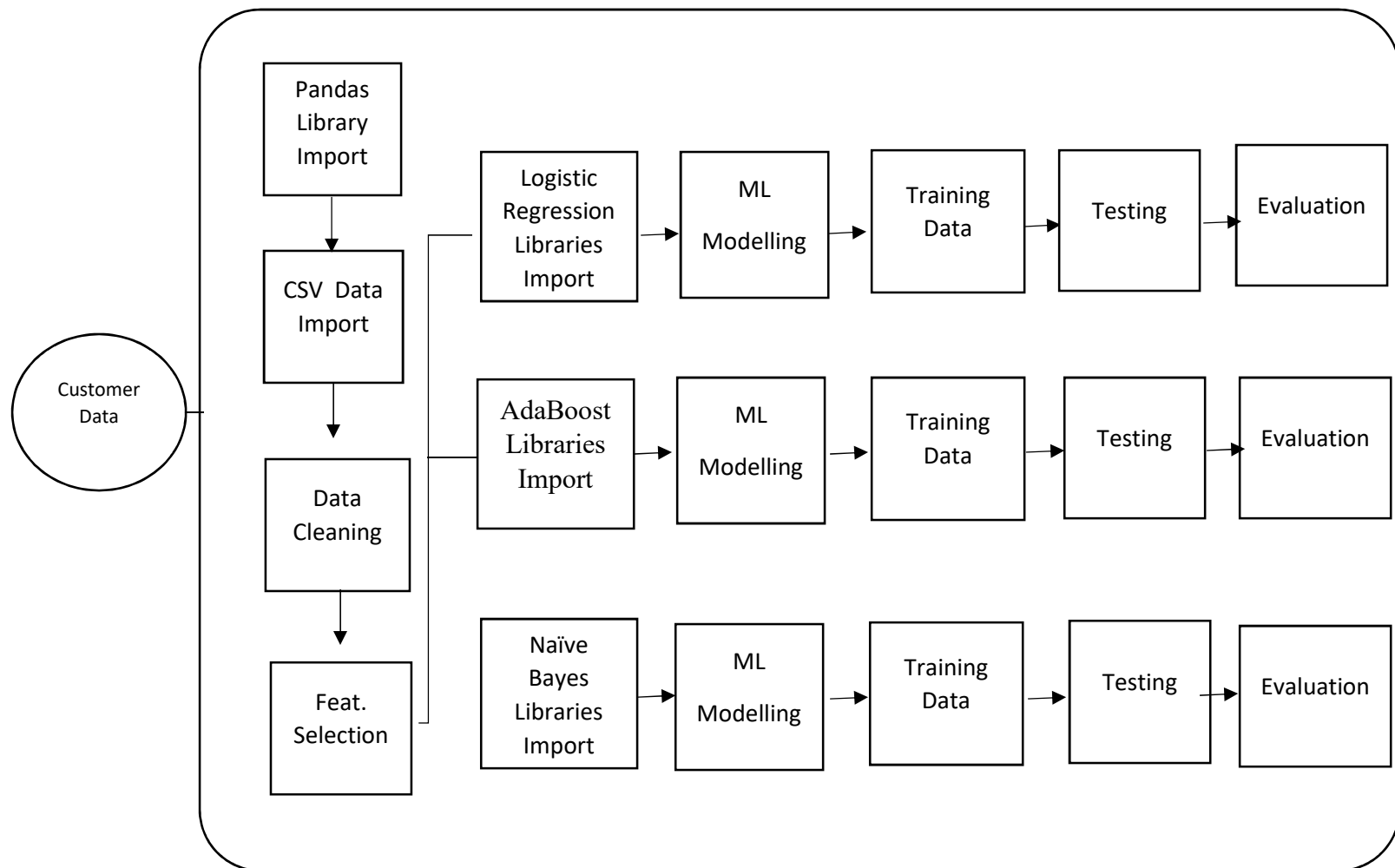


Figure 3.1: Conceptual Framework

The research methodology used to develop an ML model that can best identify customers that are likely to churn from the customer dataset of the telecom industry as expressed in the proposed conceptual framework. Figure 3.1 depicts the complete procedure of how this study was carried out.

3.1 Features of the dataset

Features such as the ones stated below were captured and extracted. Few are listed below;

- state: the state of the customer
- account_length: the length of the account
- international_plan: the international plan of the customer
- area_code: area code of the customer
- voice_mail: voicemail of the customer
- no_voicemail_msgs: where a customer has no messages
- total_day_minutes: total minutes of calls in the day
- total_day_calls: totals calls I
- total_day_charge: the total charge of calls in the day
- total_evening_minutes: the total minutes of calls during the evening
- total_night_calls: total calls in the night
- total_night_charge: total night class in the night
- total_intl_minutes: total minutes of calls during the nights
- total_intl_calls: total international calls made
- total_intl_charge: the total charge of international calls

- customer: the individual subscriber
- churn: the customer's ability to leave or stay with the network.

3.2 Classification Process

The data classification process entails data gathering and cleaning, classification and algorithm training.

Major Steps in the classification process

1. The data is categorized into a given set of classes or categories (churn). The class is also the target or label (TRUE, FALSE)
2. The classifier (the algorithm) is used to map the input data to a specific target(classes)
3. The model will be used to train the input data given for prediction and the class or category is predicted for the data once the training set is complete
4. Each sample will be assigned to one target since it is a multiclass classification
5. The model is then initialized to assign the classifier the dataset to be used for the classification
6. The classifier is trained with the `.fit(x,y)` method to fit the model for training
7. The target is then predicted for the unlabeled target
8. The model is evaluated

3.5 Performance Metrics

There are numerous metrics that can be used to assess the performance of ML algorithms, classification algorithms, and regression algorithms. All machine learning models, whether linear regression or a SOTA technique like BERT (Lin et al.,2020), require a metric to assess

performance. The metrics for evaluating ML performance is carefully chosen because the metric will help determine how the performance of ML algorithms is measured and compared. The metric chosen will completely influence the weight and the importance of the various characteristics in the result. There are dozens of performance metrics in both regression and classification but for this study, classification metrics would be the focus.

3.5.1 Classification Metrics

Confusion Matrix ($N \times N$): It is the simplest technique to evaluate a classification model whose output can include two or more different types of classes. The confusion-matrix is often used in evaluating classification problems (Gong, 2021). A confusion matrix is nothing more than a table having two dimensions, Actual and Predicted, as well as True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) for each dimension, as shown below. The `confusion_matrix` function of `sklearn.metrics` will be used to compute Confusion Matrix of the classification model under study.

		Prediction	
		Positive	Negative
Actual	Positive	TP	FN
	Negative	FP	TN

Figure 3.2: A Sample of a Confusion Matrix

True Positive (TP): the number of positive class samples the model predicts accurately

True Negative (TN): denotes the number of negative class samples that the model predicted properly

False Positive (FP): the number of negative class samples the model predicted wrongly

False Negative (FN): The number of positive class samples the model predicted wrongly

Accuracy: It is the most often used performance metric for classification algorithms -Catal (2012). It can be defined as the proportion of right predictions made to total forecasts made. We may easily calculate it using the confusion matrix and the following formula. This could be achieved by importing accuracy score metric in scikit-learn:

from sklearn.metrics import accuracy_score

$$Accuracy = \frac{\text{Number of correct predictions}}{\text{Number of All tested samples}}$$

Precision: It is the ratio of true positives and total positives predicted. $0 < P < 1$. According to Catal (2012), the precision metric is concerned with Type-I errors (FP). When a correct prediction is rejected a correct null Hypothesis (H^0), a Type-I error is made. As a result, Type-I error is incorrectly classifying samples as false positives. A model which produces no false positive has a precision of 1.0.

$$Precision = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall/Sensitivity/Hit-Rate: Basically, the ratio of true positives to all true positives in the sample. A model which produces no false negative has recall of 1.0.

$$\text{Recall/ Sensitivity (Sn)} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Specificity: The number of negatives returned by the ML model. It is the inverse of the recall metric.

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

F1-Score: The combination of precision and recall. A perfect model achieves a recall of 1.0 F1-Score.

$$\text{F1-Score} = 2 * \frac{(\text{Precision} * \text{Sensitivity})}{(\text{Precision} + \text{Sensitivity})}$$

Support: The number of samples each metric is calculated.

Macro Avg: Short for Macro Average. It is the average precision, recall and F1-Score between classes - Borgalli & Surve (2022). Macro Avg hardly class imbalance (disparities among classes) into effort, it is therefore an important metric when there are class imbalances.

Weighted Avg: Also, weighted average. It is the weighted average precision, recall and F1-Score between classes (Chaturvedi et al.,2020). It is calculated with respect to how many samples there are in each class.

3.6 Cross Validation

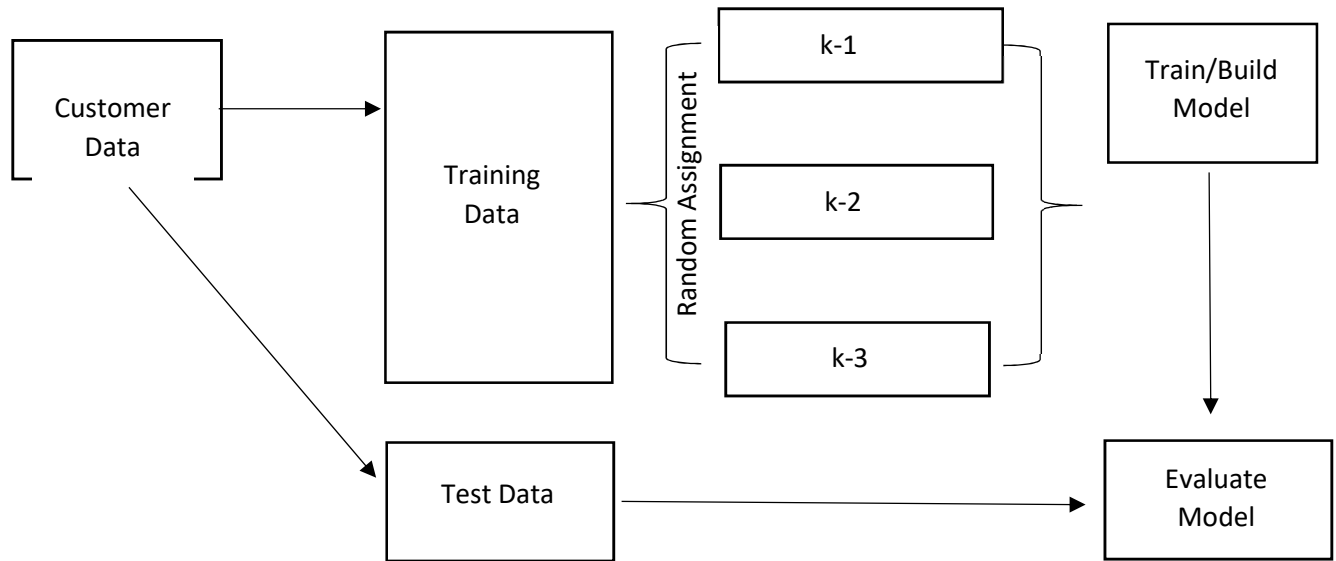


Figure 3.3: Cross Validation Framework

With K-fold Cross-validation, the set of data is divided into a K number of folds. This is used to measure the model's ability when presented with random data. The constant K denotes the number of groups into which a sample of the data is divided. With this dataset, K-value is chosen to be 10 and it is attributed to the size dataset. Each fold is utilized as a testing set. To describe the model's skill, average the values provided in step 5.

- The data is divided into training data and test data
- The k-value is selected.
- Dataset is divided into the specified number of folds (10-folds).
- Validate the model on the test dataset after training it on the training dataset.
- Save the validation result.
- Steps 1–10 are repeated, but the value of k test dataset remains random.
- The cross_val_score of the 10-folds are generated and the mean is recorded.

3.7 Experimentation Design

Scikit-learn, a Python library for machine learning, is used to carry out the experiments. The software environment is Jupyter Notebooks 3.8. All experiments are performed on an HP Pro Book with 8GB RAM and an Intel Core i5 processor running at 3.5GHz. It is powered by Windows 10. The time required to train the dataset depends on the type of system as well as the size of data collected for testing the model. In this study, 30% percent of the dataset will be used for training and 70% for testing. The percentage values would be altered to assess the performance of the sample size on the testing and training models. A supervised machine learning model is utilized with Logistic Regression, AdaBoost and Support Vector Matrix Algorithms are used for the model. A supervised model is employed in the machine learning process. The data was cleaned thoroughly since some machine learning algorithms do not support null values as attributes. The target classes were converted into a float data type in order to train and test the data set for smooth prediction.

CHAPTER 4

RESULTS AND DISCUSSION

4.0 Introduction

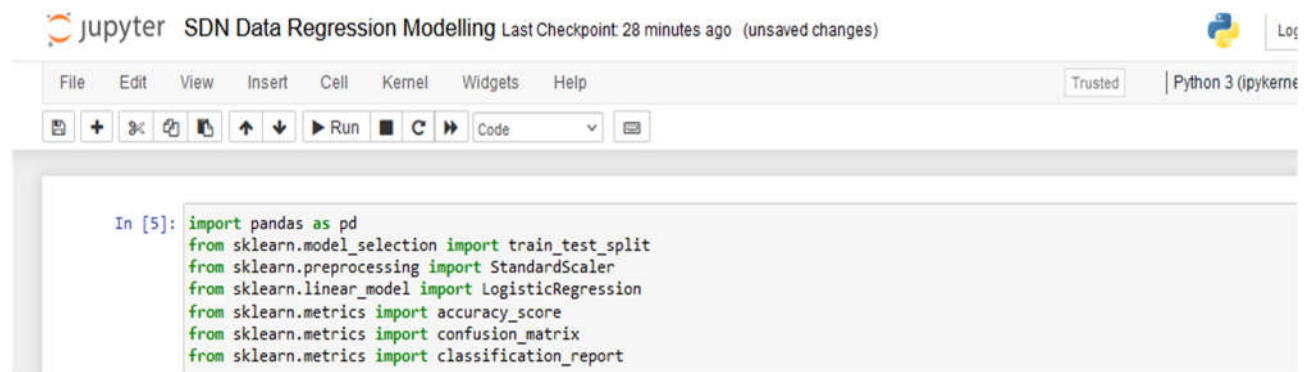
In this chapter, the results of the tests, as well as the performance of each of the selected machine learning model classifiers, are presented and analyzed to decide which has the best performance in traffic classification to upsurge the control plane's scalability performance. This section is divided into three parts. The first concerns the procedures used to prepare and feed the ML models with data acquired for analysis. In the second part, the outcomes of tests on all three Machine Learning algorithms employed on the training and testing sets with confusion matrix are analyzed.

4.1 ML Modelling and design

This section discusses how the customer data classifier's training data is collected. Class 1, Class 2, and Class 3 customer data were imported and prepared for usage with machine learning models using python scripts. It began with the pip install command, which installed the libraries required to run the code, namely numpy, pandas, seaborn, and scikit-learn. The script folder was found in the C-programs drive's folder, where the Python program's files were installed. The following commands are entered into the command line interface (CLI) after launching. Some of the libraries had to be downloaded and installed in real-time.

```
C:\Users\Administrator\AppData\Local\Programs\Python\Python38\Scripts>pip install scikit-learn
Collecting scikit-learn
  Downloading scikit-learn-1.1.3-cp38-cp38-win_amd64.whl (7.5 MB)
    | 7.5 MB 1.7 MB/s
Requirement already satisfied: numpy>=1.17.3 in c:\users\administrator\appdata\local\programs\python\python38\lib\site-packages (from scikit-learn) (1.23.4)
Collecting scipy>=1.3.2
  Downloading scipy-1.9.3-cp38-cp38-win_amd64.whl (39.8 MB)
    | 39.8 MB 22 kB/s
Collecting threadpoolctl>=2.0.0
  Downloading threadpoolctl-3.1.0-py3-none-any.whl (14 kB)
Collecting joblib>=1.0.0
  Downloading joblib-1.2.0-py3-none-any.whl (297 kB)
    | 297 kB 1.3 MB/s
Installing collected packages: scipy, threadpoolctl, joblib, scikit-learn
Successfully installed joblib-1.2.0 scikit-learn-1.1.3 scipy-1.9.3 threadpoolctl-3.1.0
WARNING: You are using pip version 20.2.3; however, version 22.3.1 is available.
You should consider upgrading via the 'c:\users\administrator\appdata\local\programs\python\python38\python.exe -m pip install --upgrade pip' command.
```

Important libraries to the study were also imported including `train_test_split`, `StandardScaler` as well as `classification_report`. All these were done in the Jupyter python notebook environment. The environment is live and runs in a browser.



```
In [5]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
```

Uploading, exploring and cleaning data: The `dataset_sdn.csv` file was uploaded as a result of an earlier collection from the simulations of the controller into a `.py` script. The dataset was cleaned considering Logistic Regression Model's inability to work on NaN tuples.

The script '`isnull().values.any()`' was used to query the dataset. The result indicated a true value which meant there were tuples that had NaN values. The cleaning ended with an excel formula removing all the NaN tuples of each attribute from the dataset. This was necessary as result of the dataset size. The action was repeated until the script returned a false value indicating a zero NaN tuple.

```
In [25]: #Data cleaning and validation using isnull
data.isnull().values.any()

Out[25]: False

In [24]: nan = data.isnull()
print(nan)
```

Data Description: The `data.info()` script was used and the dataset had 5,000 customers.

To remove target and ID columns because the arbitrary ID data isn't relevant to the analysis

```
features = customers[customers.columns.difference(['area_code', 'phone_number', 'class'])]
```

```
In [10]: x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.2, random_state =42)
        scaler = StandardScaler()
        scale = scaler.fit(x_train)
        x_train =scale.transform(x_train)
        x_test =scale.transform(x_test)
```

Fitting the model: The Logistic Regression model was fitted on the trained set and the model predicted based on the trained sets. The ‘model.predict(x_test)’ was used to predict the corresponding values.

```
In [33]: #Creating a Logistic Regression Model
        model =LogisticRegression()
        model.fit(x_train,y_train)
        pred=model.predict(x_test)

c:\users\administrator\appdata\local\programs\python\python38\lib\site-packages\sklearn\
vergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
```

4.2 Performance Analysis

The usage and implementation of the performance metric for each of the machine learning models are briefly explained in this section.

4.2.1 Confusion matrix

The confusion matrix table was utilized to evaluate the efficacy of the techniques put in place following the categorization. It served as a tool for graphically illustrating the numerical breakdown of the model's predictions by class. The predictions of the machine learning model are compared in a matrix against the actual target values. This provides a thorough grasp of the

efficacy of each categorization model as well as the kinds of errors it generates. By providing real and predicted values to the function, the confusion matrix for each ML algorithm was derived. Accuracy (A), precision (P), sensitivity (Sn), and F1-Score values were the performance measurements for each of the three algorithms that were categorized using these characteristics. For a multi-class classification problem like the one being studied, accuracy is defined as the weighted correct classifications across both outputs. The correctness of the test is gauged by the F1-Score. Also demonstrated are the sensitivity and precision. Precision is the ratio of the number of relevant examples among the instances that were retrieved to the total number of relevant examples.

4.2.2 Logistic Regression Analysis

Regression approaches can measure associations, forecast outcomes, and correct for confounding variable effects, making them useful in a variety of research applications. One such method is logistic regression, which measures the individual contribution of each independent variable to the overall influence of a set of independent variables on a binary result (Stoltzfus, 2011).

4.2.2.1 Confusion matrix: It is the performance of the classification models for the dataset after prediction and it is evaluated. This can only be determined if the true values for test data are known.

```
In [35]: #creating a confusion matrix
         confusion_matrix(y_test,pred)

Out[35]: array([[3427,  0,  0],
                [ 0, 2931, 12],
                [ 4, 19, 4042]], dtype=int64)
```

	Class 1	Class 2	Class 3
Class 1	342	0	0
Class 2	0	2931	12
Class 3	4	19	4042

Figure 4.1: Logistic Regression Confusion Matrix

Accuracy: This metric measure how many forecasts made using the predictive model were accurate in comparison to all other predictions. Per the model's prediction, accuracy(A)

$$A = 0.9966$$

$$\text{Accuracy} = 99.66\%$$

```
In [34]: score = accuracy_score(y_test, pred)
          score
Out[34]: 0.9966459032103497
```

Precision: It is the proportion of all predictions made by the predictive model that turned out to be true. From the classification report, a precision of 1.00, 0.99 and 1.00 were obtained by the model. The average is then computed and hence the result.

$$\text{Precision(P)} = \text{Average} (1.00 + 0.99 + 1.00)$$

$$\text{Precision(P)} = 0.9966 (99.67\%)$$

```
In [38]: # Confusion Matrix_METRICS
matrix=classification_report(y_test,pred)
print('Classification report : \n', matrix)
```

```
Classification report :
              precision    recall  f1-score   support

     1         1.00      1.00      1.00     3427
     2         0.99      1.00      0.99     2943
     3         1.00      0.99      1.00     4065

 accuracy          1.00          1.00          1.00     10435
 macro avg          1.00          1.00          1.00     10435
 weighted avg          1.00          1.00          1.00     10435
```

Recall/Sensitivity: This is a metric that represents the percentage of traffic that was actually the same as predicted. This addresses the issue of the model's inability to distinguish good instances.

From the classification report, a precision of 1.00, 1.00 and 0.99 were obtained by the model.

The average of the individual classes is then computed and hence the result.

$$\text{Sensitivity (Sn)} = \text{Average } (1.00 + 1.00 + 0.99)$$

$$\text{Sensitivity (Sn)} = 0.9966 \text{ (99.67\%)}$$

F1 – Score: It speaks of the recall and precision harmonic means. It allows a model with high recall but low precision, or the opposite, to be comparable.

$$F1\text{-Score} = 2 * \frac{(\text{Precision} * \text{Sensitivity})}{(\text{Precision} + \text{Sensitivity})}$$

$$F1\text{-Score} = 2 * \frac{(0.9966 * 0.9966)}{(0.9966 + 0.9966)}$$

$$= 0.9966$$

$$= 99.66\%$$

The higher a model's F-Score, the more accurate it is. The lower a model's F-Score, the less accurate it is. This model performed reasonably well as a classifier for the dataset, having a relatively high F-Score and a higher precision score.

<i>Metrics</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Sensitivity/Recall</i>	<i>F1-Score</i>
Logistic Regression	99.66%	99.67%	99.67%	99.66%

Table 1: Performance of Logistic Regression

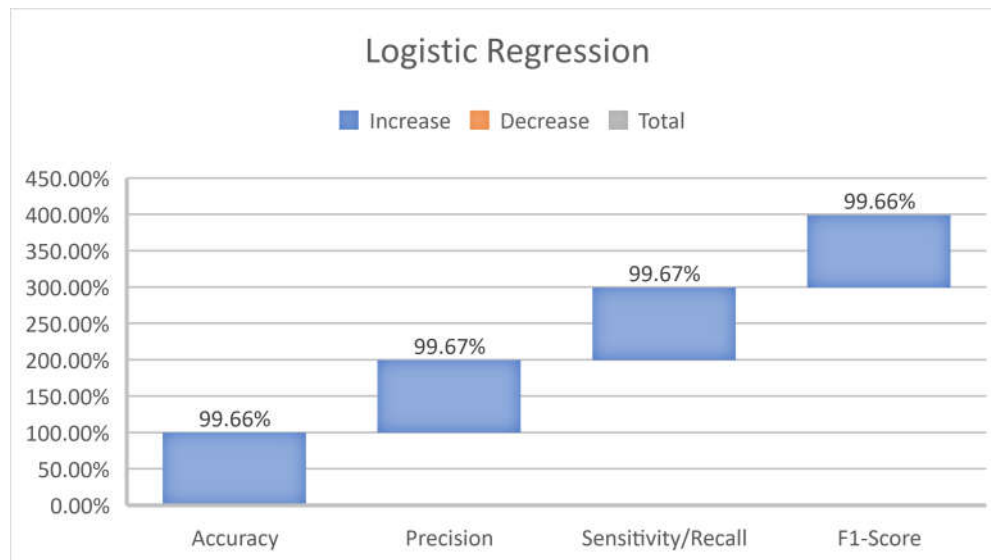


Figure 4.2: Performance of Logistic Regression

4.2.3 AdaBoost's Analysis

According to Peterson (2009), when there is little or no prior knowledge about the distribution of the data, AdaBoost classification is one of the first options considered. A simple but effective machine learning algorithm. Both classification and regression can be done successfully using it.

However, classification prediction is where it is most frequently utilized. The AdaBoost identifies newly input data based on its similarity to previously trained data and organizes the data into coherent clusters or subsets. The class to which the input belongs is determined by its closest class. Figure 4.3: AdaBoost

4.2.3.1 Confusion matrix:

	Class A	Class B	Class C
Class A	10038	0	0
Class B	1	8903	40
Class C	1	19	12302

Figure 4.3: AdaBoost Confusion Matrix

Accuracy:

```
In [68]: score = accuracy_score(y_test, pred)
          score
Out[68]: 0.9980513672374137
```

Accuracy(A): 0.9981(99.81%)

```
In [71]: # Confusion Matrix_METRICS
matrix=classification_report(y_test,pred)
print('Classification report : \n', matrix)
```

```
Classification report :
              precision    recall  f1-score   support

     1           1.00       1.00       1.00       10038
     2           1.00       1.00       1.00        8944
     3           1.00       1.00       1.00       12322

 accuracy              1.00       1.00       1.00       31304
 macro avg           1.00       1.00       1.00       31304
 weighted avg        1.00       1.00       1.00       31304
```

From the classification report above, the precision, recall, F1-Score as well as support are as follows.

Precision(P):

$$\text{Precision(P)} = \text{Average } (1.00 + 1.00 + 1.00)$$

$$\text{Precision(P)} = 1.00 \text{ (100.00\%)}$$

Sensitivity/Recall

$$\text{Sensitivity (Sn)} = \text{Average } (1.00 + 1.00 + 1.00)$$

$$\text{Sensitivity (Sn)} = 1.00 \text{ (100.00\%)}$$

F1-Score

$$\text{F1-Score} = \text{Average } (1.0 + 1.0 + 1.0)$$

$$= 1.00 \text{ (100\%)}$$

<i>Metrics</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Sensitivity/Recall</i>	<i>F1-Score</i>
AdaBoost	99.81%	100%	100%	100%

Table 2: Performance of AdaBoost

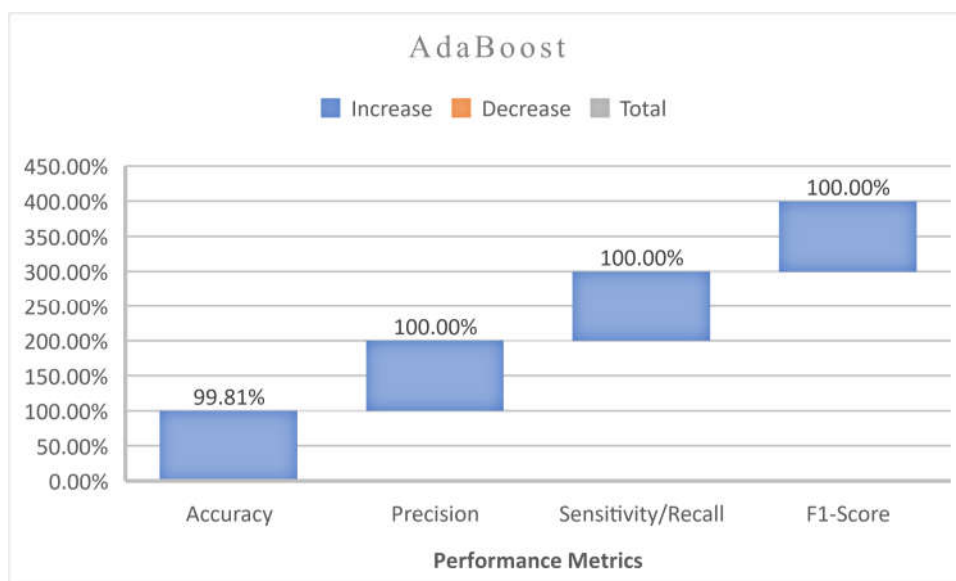


Figure 4.4: Performance of AdaBoost

4.2.4 Gaussian's Naïve Bayes

It is a classification technique built on the Bayes Theorem and predicates on the idea of predictor independence. To put it simply, a Naives Bayes classifier believes that the presence of one feature in a class has nothing to do with the presence of any other feature. The libraries are imported.

```
from sklearn.naive_bayes import GaussianNB
```

```
In [9]: x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.3, random_state =42)
        scaler = StandardScaler()
        scale = scaler.fit(x_train)
        x_train =scale.transform(x_train)
        x_test =scale.transform(x_test)
```

```
In [21]: #Creating a Bayes Model
        # p represents Euclidian Distance
        Bayes =GaussianNB()
        Bayes.fit(x_train,y_train)

        #Predict the value for new unseen data
        Bayes_pred= Bayes.predict(x_test)
        Bayes_pred
```

```
Out[21]: array([2, 2, 1, ..., 2, 1, 3], dtype=int64)
```

4.2.4.1 Confusion matrix:

```
In [23]: #creating a confusion matrix
        confusion_matrix(y_test,Bayes_pred)
```

```
Out[23]: array([[10038,    0,    0],
               [    0, 8944,    0],
               [  722, 5912, 5688]], dtype=int64)
```

	Class A	Class B	Class C
Class A	10038	0	0
Class B	1	8944	40
Class C	722	5912	5688

Figure 4.5: Naives Bayes Confusion Matrix

```
In [24]: #Accuracy
Accuracy = accuracy_score(y_test, Bayes_pred)
Accuracy

Out[24]: 0.7880782008688986
```

Accuracy = 0.7881(78.81%)

```
In [22]: print(classification_report(y_test, Bayes_pred))
```

	precision	recall	f1-score	support
1	0.93	1.00	0.97	10038
2	0.60	1.00	0.75	8944
3	1.00	0.46	0.63	12322
accuracy			0.79	31304
macro avg	0.84	0.82	0.78	31304
weighted avg	0.86	0.79	0.77	31304

Precision(P):

Precision(P) = Average (0.93+ 1.00 + 0.60)

Precision(P) = 0.84 (84.00%)

Sensitivity/Recall

Sensitivity (Sn) = Average (1.00 + 1.00 + 0.46)

Sensitivity (Sn) = 0.82 (82.00%)

F1-Score

$$F1\text{-Score} = \text{Average } (0.97 + 0.75 + 0.63)$$

$$= 0.78 \text{ (78.00\%)}$$

<i>Metrics</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Sensitivity/Recall</i>	<i>F1-Score</i>
Naive Bayes	78.81%	84.00%	82.00%	78.00%

Table 3: Performance of Naïve Bayes Classifier

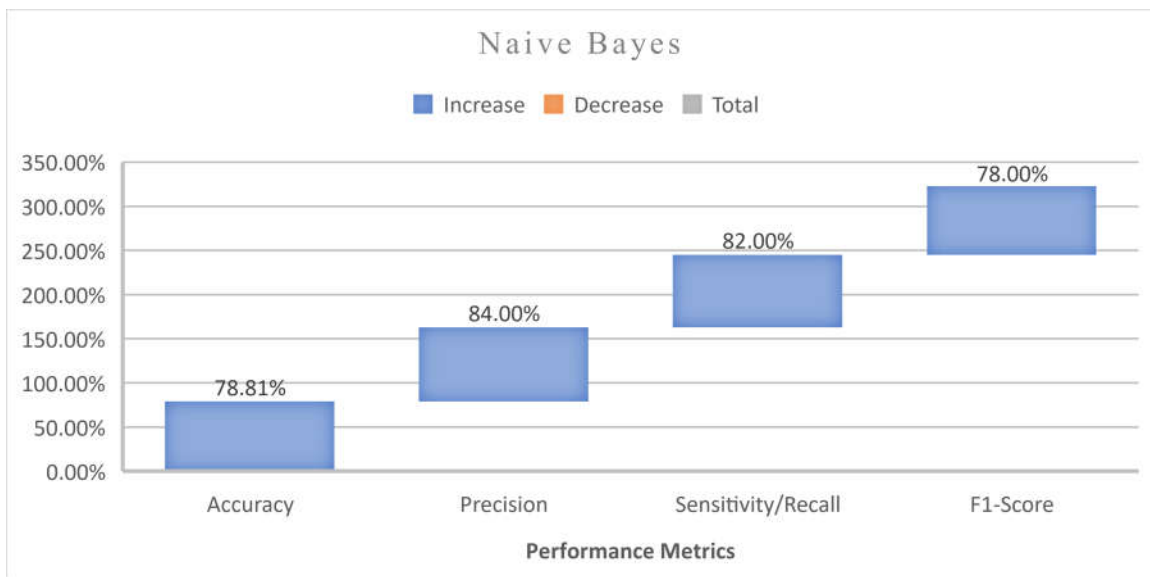


Figure 4.6: Performance of Naïve Bayes

The table below indicates the performance of the various algorithms and their performance on dataset in the experimentation. (Logistic Regression, AdaBoost and Naïve Bayes)

<i>Performance Metric</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Sensitivity/Recall</i>	<i>F1-Score</i>
Logistic Regression	99.66%	99.67%	99.67%	99.66%
AdaBoost	99.81%	100%	100%	100%
Naive Bayes	78.81%	84.00%	82.00%	78.00%

Table 4: Confusion Matrix Performance Table

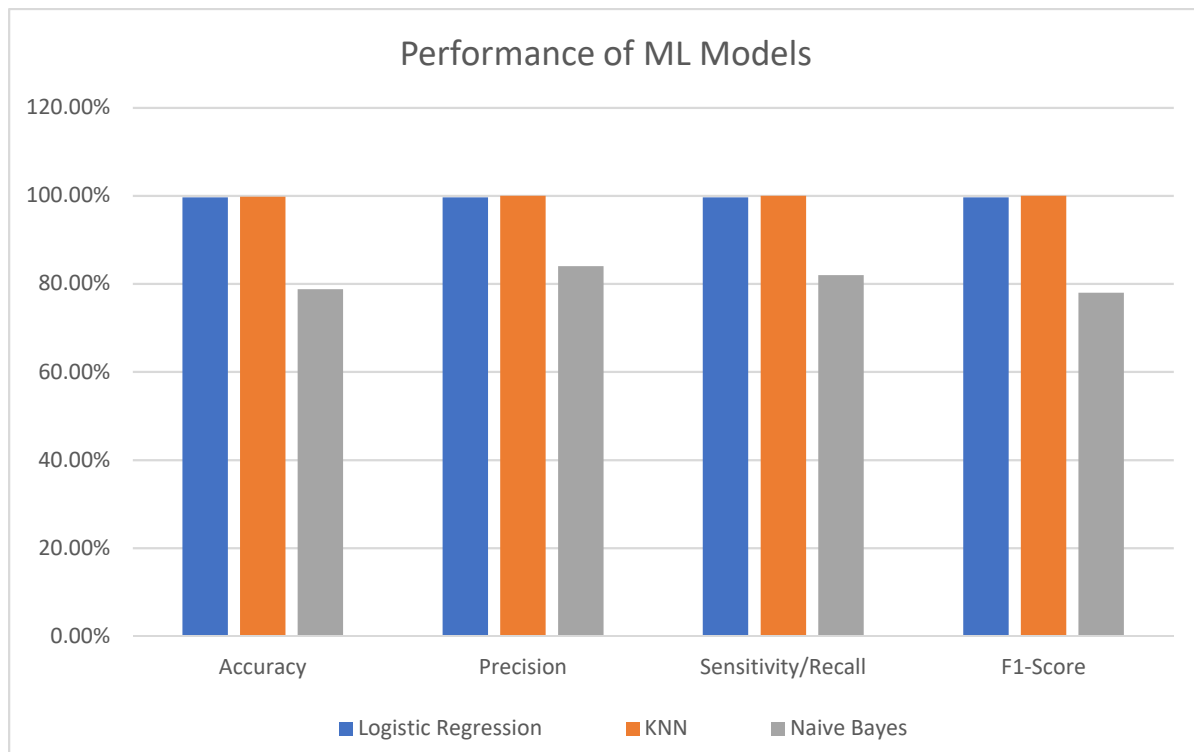


Figure 4.7: Performance Metrics on the ML Models

4.3 Cross Validation

4.3.1 Logistic Regression Cross Validation

Using K-fold Cross Validation, where the value of K=10. The dataset was divided into 80% and 20% for Training and Testing respectively. The results are as follows

The cross_val_score function is imported and the cross-validation value of 10 is scripted.

```
In [25]: from sklearn.model_selection import cross_val_score
scores= cross_val_score(mymodel,x_train,y_train,scoring='r2', cv =10)

In [26]: scores
Out[26]: array([0.99357088, 0.99416894, 0.99566408, 0.99147768, 0.99297283,
0.99671068, 0.99372068, 0.99491587, 0.9937196 , 0.99476633])

In [31]: print((scores.mean(), scores.std()))
(0.9941687577936174, 0.0013800280076544454)
```

4.3.2 AdaBoost Cross Validation

The result of the cross validation is presented

```
In [33]: scores
Out[33]: array([0.99578087, 0.9944375 , 0.99789043, 0.99769865, 0.99712331,
0.99673976, 0.99712331, 0.99750687, 0.9963562 , 0.99750687])

In [34]: print((scores.mean(), scores.std()))
(0.9968163774702516, 0.0010011475252408822)
```

4.3.3 Naïve Bayes Cross Validation

```
In [19]: from sklearn.naive_bayes import GaussianNB

In [20]: Bayes=GaussianNB()
Bayes_model =Bayes.fit(x_train,y_train)

In [22]: from sklearn.model_selection import cross_val_score
scores= cross_val_score(Bayes_model,x_train,y_train,scoring='r2', cv =10)
```

The result of the cross validation is presented

```

In [23]: scores
Out[23]: array([0.5847988 , 0.6071724 , 0.59285308, 0.59899001, 0.59592154,
                0.59381197, 0.58422302, 0.58307235, 0.58307235, 0.59035995])

In [24]: print((scores.mean(), scores.std()))
          (0.591427548621293, 0.007547243886535934)

```

<i>Metric</i>	<i>Cross Validation Score</i>	<i>Standard Deviation</i>
Logistic Regression	99.42%	0.0014
AdaBoost	99.68%	0.0010
Naïve Bayes	59.14%	0.0075

Table 5: Cross Validation Table

CHAPTER 5

SUMMARY, CONCLUSION AND RECOMMENDATION

5.0 Introduction

The telecommunications sector has experienced significant loss due to high churn rates. While business losses are inevitable, turnover can still be controlled and maintained at a respectable level. To avoid issues for the telecommunications business, good techniques must be established and existing ones must be improved. In this study, we examined and explored three distinct prediction models, including AdaBoost and Logistic Regression, as well as their performance metrics. AdaBoost is definitely an efficient technique, as evidenced by the fact that the accuracy they obtain is significantly higher than that of the logistic regression technique. In this chapter, an overview of the work and the findings of the study are presented. The study is summarized in Section 5.1, which also covers the inferences drawn from the findings of Chapter 4. The study's recommendations for future work are discussed in Section 5.2.

5.1 Summary of the Research

This research conducted an experimental investigation of customer churn prediction based on real-date sets. The research has addressed the problem of uncertain customers with the highest churn propensity and retention mechanisms shifted in place. The difference is the way in which this research has proven the AdaBoost algorithm as the best performing classifier on the telecom data set. It has become extremely important and necessary to choose the best machine learning classifier for the telecom churn data classification and, as a result, effectively predicted customers who are likely to churn. This would help improve on the decision making process for players in

the telecom industry as wrong decisions could have high negative financial implications on businesses and products. The telecom churn problem is addressed in this thesis by a comparative analysis of the effectiveness of a few chosen machine learning classification algorithms. Although the Logistic Regression Algorithm, and Naive Bayes Machine learning classifiers were chosen in addition to AdaBoost. The algorithms were selected based on the findings of other researchers. For instance, Naïve Bayes was mentioned by Peterson (2009) as the first classifier to be considered in a classification task. A Bayesian Machine Learning Algorithm stood out for Flow Prediction (Baz ,2018). Using a Jupyter notebook and customer data collected from a range of customers, the experiment was carried out. When recognizing the right customer data class (Class A, B and C) were used from the dataset, performance measurements like the Confusion Matrix and Cross Validation metrics are utilized to compare different machine learning algorithms. Accuracy, Precision, Sensitivity (Recall), and f-1 Score are the four major criteria used to compare the performance of ML classifiers. For enhanced visual comprehension, histograms, and a radar chart are used to show which aspects of the datasets performed well or poorly as well as their capacity to fit or overfit the models. The most effective ML classifier for categorizing traffic data in the dataset is that which performs the best based on these criteria. The accuracy of the predictions made by AdaBoost, Logistic Regression, and Naive Bayes from the classes of the datasets were 99.81%, 99.66%, and 78.81%, respectively. This therefore answers the question; Which machine learning techniques are used in evaluating customer churn prediction? According to additional test findings (*figure 4.7*) in terms of precision, sensitivity, and f1-score, AdaBoost outperformed the other two classifiers with a 100% score. Naive Bayes had poor results with the customer datasets and is therefore an unsuitable classifier for telecom data classification' in real-time.

With the k-fold cross validation, AdaBoost maintained an excellent score of 99.68 percent for the cross-validation results, while Logistic Regression scored 99.42 percent. The original 78.81 percent performance by Naive Bayes dropped further to 59.14 percent. However, in the case of AdaBoost, trade-off between precision and recall prevents it from being able to maximize both at the same time in practice.

In order to answer the question, is there a machine learning model that is most capable of identifying customers churn in the telecom sector? On this dataset, which included voice mail plan, customer type, AdaBoost performed admirably well for both confusion matrix and cross validation measures. It is worth noting that, one machine learning algorithm cannot be declared superior to another, as the characteristics of a specific machine learning technique on a particular dataset may not synchronize correctly and could affect performance. An ML algorithm's several steps such as abstraction, cleaning, categorization, and tuning, may also affect how well an algorithm performs.

5.2 Future Enhancement

Datasets from various sectors can be analyzed. A heterogeneous dataset with diverse properties could be projected to perform differently on the selected classifiers. However, machine learning classifiers can be trained on new datasets from other sectors such as the banking sector, and their efficacy assessed in that area. It is evident that various datasets might perform differently on the same or different classifiers. Additionally, try using multiple Machine Learning Classifiers on different targets or attributes. To determine performance differences, different classifiers could be tested on a single-class target or category as opposed to a multi-class target and vice versa.

Machine Learning algorithms have constraints in terms of datasets that may be used, and these constraints can be addressed. To determine whether a dataset is suitable for a classifier, attributes such as type, null, and others can be investigated.

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